Similarly, most investment advisers registered with the Commission are limited geographically. These advisers must generally make a "notice filing" with a state in which they have a place of business or six or more clients.[655] Figure 7 plots the frequency distribution of the number of such filings. Based on notice filings, 57 percent of investment advisers registered with the Commission operated in fewer than four states, and 37 percent operated in only one state.[656]

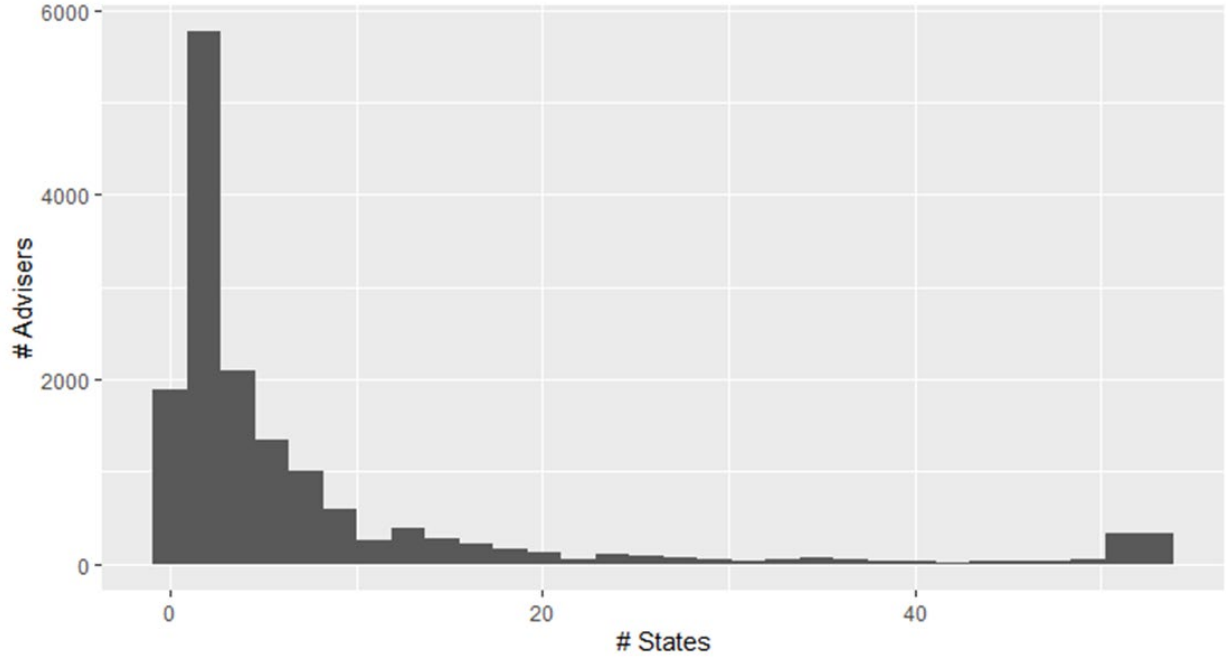

Figure 7: Frequency of number of state notice filings by SEC-registered investment advisers. Data source: Form ADV, Item 2.C (as of Oct. 5, 2023).

d. Investment Companies

Investment companies are companies that issue securities and are primarily engaged in the business of investing in securities. Investment companies invest money they receive from investors on a collective basis, and each investor shares in the profits and losses in proportion to that investor's interest in the investment company. Investment companies subject to the final

655 *See* General Instructions to Form ADV (as of Oct. 5, 2023).

656 Form ADV, Item 2.C (as of Oct. 5, 2023). This includes 1,887 advisers who do not make any notice filings.

amendments include registered open-end and closed-end funds, business development companies ("BDCs"), Unit Investment Trusts ("UITs"), employee securities' companies ("ESCs"), and management company separate accounts ("MCSAs"). Because they are not operating companies, investment companies do not have "customers" as such, and thus are unlikely to possess significant amounts of nonpublic "customer" information in the conventional sense. They may, however, have access to nonpublic information about their investors.[657]

Table 4 summarizes the investment company universe that will be subject to the final amendments. In total, as of September 30, 2023, there were 13,766 investment companies, including 12,183 open-end management investment companies, 682 closed-end managed investment companies, 702 UITs,[658] 141 BDCs,[659] approximately 43 ESCs, and 15 MCSAs. Many of the investment companies that will be subject to the final amendments are part of a "family" of investment companies.[660] Such families often share infrastructure for operations (e.g., accounting, auditing, custody, legal), and potentially marketing and distribution. We expect that many of the compliance costs and other economic costs discussed in the following sections will likely be borne at the family level.[661] We estimate that there were up to 1,131 distinct

---

657 The definition of "customer information" in the final amendments includes information about investment companies' investors. *See* final rule 248.30(d)(5)(i) and § 248.3(t).

658 For this release, the number of UITs includes N-4, N-6, N-8B-2, and S-6 filers as of Sept. 30, 2023. The Proposing Release cited a figure of 662 UITs using 2021 N-CEN filings. The correct number of UITs using 2021 N-CEN filings in the Proposing Release should be 703. This change would not have affected the Commission's assessment of economic effects at Proposal as these assessments were focused primarily on effects at the level of individual covered institutions and their customers.

659 For this release, the number of BDCs was estimated using London Stock Exchange Group ("LSEG") BDC Collateral data as of Sept. 2023.

660 As used here, "family" refers to a set of funds reporting the same family investment company name (Form N-CEN Item B.5) or filing under the same registrant name (Form N-CEN Item B.1.A).

661 For example, each investment company in a family is likely to share common policies and procedures.

operational entities (families and unaffiliated investment companies) in the investment company universe.[662]

Table 4: Investment Companies, summary statistics. For each type of fund, this table presents estimates of the number of investment companies and investment company families. Data sources: 2023 N-CEN filings,[a] LSEG BDC Collateral (2023).[b]

Fund Type	# Inv. Co.	# Families[c]	# Unaffiliated[d]	# Entities[e]
Open-End[f]	12,183	212	251	463
Closed-End[g]	682	87	153	240
UIT[h]	702	106	229	335
BDC[i]	141	-	-	141
ESC[j]	43	-	-	43
MCSA[k]	15	1	1	2
Total[l]	13,766	313	634	1,131

a Year 2023 Form N-CEN (as of Sept. 2023).
b LSEG BDC Collateral (as of Sept. 2023).
c Number of families calculated from affiliation reported by registrants on Item B.5 of form N-CEN. The total number of families represents the number of distinct families; summing over the number of families across different fund types will double count some fund families.
d Number of registrants reporting no family affiliation.
e Number of distinct entities, i.e., the sum of distinct families (# Families) and unaffiliated registrants (# Unaffiliated). The grand total is the sum of distinct families (313), total unaffiliated registrants (634), BDCs (141), and ESCs (43).
f Form N-1A filers; includes all open-end funds, including ETFs registered on Form N-1A.
g Form N-2 filers not classified as BDCs.
h UITs are comprised of (1) Variable annuity separate accounts organized as UITs, which are series, or classes of series, of trusts registered on Form N-4; (2) Variable life insurance separate accounts organized as UITs, which are series, or classes of series, of trusts registered on Form N-6; (3) ETFs organized as UITs, which are series, or classes of series, of trusts registered on Form N-8B-2 / S-6 (Non-separate-account UITs register in the first instance on form N-8B-2, and then their subsequent filings are on Form S-6); and Non-ETF UITs are trusts registered on Forms N-4 or N-6.
i Form N-2 filers classified as BDCs.
j Form 40-APP filers [not classified as BDCs].
k Trusts registered on Form N-3.
l Cells do not sum to totals as investment company families may span multiple investment company types.

662 For this release, the number of unaffiliated entities was estimated using N-CEN filings as of Sept. 30, 2023. The Proposing Release cited a figure of 476 using 2021 N-CEN filings. The correct number of the unaffiliated entities using 2021 N-CEN filings in the Proposing Release should be 609. This change would not have affected the Commission's assessment of economic effects at Proposal as these assessments were focused primarily on effects at the level of individual covered institutions and their customers.

e. Transfer Agents

Transfer agents maintain records of security ownership and are responsible for processing changes of ownership ("transfers"), communicating information from the firm to its security-holders (e.g., sending annual reports), replacing lost stock certificates, etc. However, in practice, most securities registered in the U.S. are held in "street name," where the ultimate ownership information is not maintained by the transfer agent but rather in a hierarchal ledger. In this structure, securities owned by individuals are not registered in the name of the individual with the transfer agent. Rather, the individual's broker maintains the records of the individual's ownership claim on securities. Brokers, in turn, have claims on securities held by a single nominee owner who maintains records of the claims of the various brokers.[663] In such cases, the transfer agent is not aware of the ultimate owner of the securities and therefore does not hold sensitive information belonging to those owners, as only the broker holds this information.

Despite the prevalence of securities held in street name, a large number of individuals nonetheless hold securities directly through a transfer agent. Securities held directly may be held either in the form of a physical stock certificate or in book-entry form through the Direct Registration System ("DRS"). In either case, the transfer agent would need to maintain sensitive information about the individuals who own the securities. For example, to handle a request for replacement certificate, the transfer agent would need to confirm the identity of the individual making such a request and to maintain a record of such confirmation. Similarly, to effect DRS transfers, a transfer agent would need to provide a customer's identification information in the message to the DRS.

---

[663] In the U.S., this owner is generally Cede & Co., a partnership organized by the Depository Trust & Clearing Corporation.

In 2023, there were 251 transfer agents registered with the Commission, with an additional 64 registered with the Banking Agencies.[664] As discussed above,[665] differences in the baseline regulation of these transfer agents affect their current notification obligations.[666] Among the 315 transfer agents, 132 are considered small entities.[667] By registration, 100 of these small transfer agents are registered with the Commission and 32 are registered with the Banking Agencies.[668]

On average, each transfer agent reported around 1 million individual accounts, with the largest reporting 61 million.[669] Figure 8 plots the cumulative distribution of the number of individual accounts reported by registered transfer agents. Approximately one third of registered

---

664 Form TA-1 (as of Sept. 30, 2023).

665 *See supra* footnotes 601 and 604 and accompanying text.

666 *See infra* sections IV.D.2.b and IV.E (discussing benefits and costs, and competitive effects, relative to the baseline).

667 *See infra* section VI.C. Estimate based on the number of transfer agents that reported a value of fewer than 1,000 for items 4(a) and 5(a) on Form TA-2 collected by the Commission as of Sept. 30, 2023.

668 *Id*.

669 Form TA-2 Items 5(a) (as of Sept. 30, 2023). This analysis is limited to the 265 transfer agents that filed form TA-2. For the 205 transfer agents registered with the Commission that filed form TA-2, the average number of individual accounts is 1.2 million; for the 60 transfer agents registered with the Banking Agencies that filed form TA-2, the average number of individual accounts is 69 thousand.

transfer agents reported no individual accounts,[670] and 58 percent reported fewer than ten thousand individual accounts.[671]

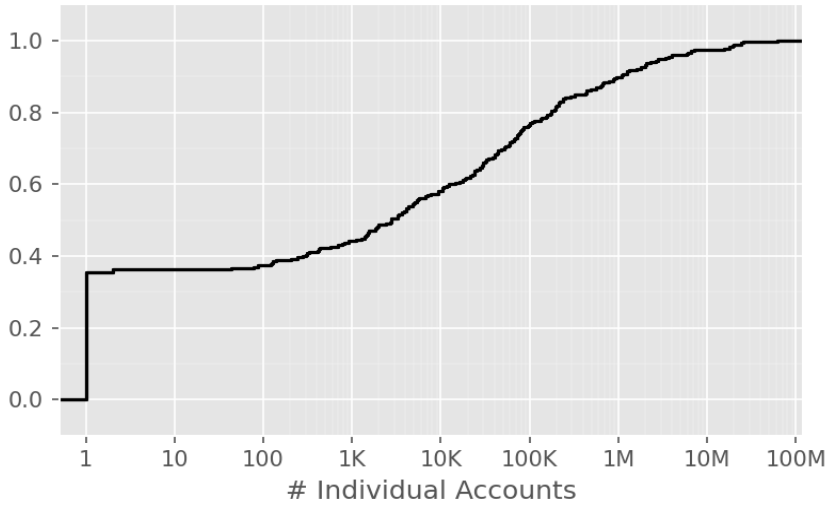

Figure 8: Cumulative distribution of the number of individual accounts (logarithmic scale) across registered transfer agents. Data source: Form TA-2, Items 5(a) (as of Sept. 30, 2023).

f. Service Providers

The final amendments require that a covered institution's incident response program include the establishment, maintenance, and enforcement of written policies and procedures reasonably designed to require oversight, including through due diligence and monitoring, of service providers. These policies and procedures must be reasonably designed to ensure service providers take appropriate measures to protect against unauthorized access to or use of customer information and to notify covered institutions of an applicable breach in security.[672] These requirements on a covered institution will affect a service provider that "receives, maintains,

670 Some registered transfer agents outsource many functions—including tracking the ownership of securities in individual accounts—to other transfer agents ("service companies"). *See* Form TA-1 Item 6 (as of June 20, 2022).

671 Form TA-2, Items 5(a) (as of Sept. 30, 2023).

672 *See* final rule 248.30(a)(5).

processes, or otherwise is permitted access to customer information through its provision of services directly to [the] covered institution."[673]

Covered institutions' relationships with a wide range of service providers will be affected. Specialized service providers with offerings geared toward outsourcing of covered institutions' core functions will generally fall under the requirements. Those offering customer relationship management, customer billing, portfolio management, customer portals (e.g., customer trading platforms), customer acquisition, tax document preparation, proxy voting, and regulatory compliance (e.g., AML/KYC) will likely fall under the requirements. Some of these specialized service providers will be themselves covered institutions.[674] In addition, various less-specialized service providers might potentially fall under the requirements. Service providers offering Software-as-a-Service (SaaS) solutions for email, file storage, and similar general-purpose services might potentially be in a position to receive, maintain, or process customer information. Similarly, providers of Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), as well as those offering more "traditional" consulting services (e.g., IT contractors) will in many cases be "otherwise [] permitted access to customer information" and might fall under the provisions.

In the Proposing Release, we stated that the financial services industry is increasingly relying on service providers through various forms of outsourcing.[675] We also stated that we were unable to quantify or characterize in much detail the structure of the relevant service provider markets due to data limitations.[676] One commenter stated that this resulted in an analysis that

---

673 Final rule 248.30(d)(10).

674 For example, many investment companies rely on third-party investment advisers and transfer agents.

675 *See* Proposing Release at section III.C.3.e; *see also* Bank for International Settlements, *Outsourcing in Financial Services* (Feb. 15, 2005), *available at* https://www.bis.org/publ/joint12.htm.

676 *See* Proposing Release at section III.C.3.e.

fails to meaningfully address the associated costs.[677] While this commenter did not identify any additional data sources, in response we have conducted a further review of industry literature.[678] While we continue to find certain data limitations, we also have identified certain additional informative data points on covered institutions' reliance on service providers.[679] A recent notice issued by FINRA states that FINRA's members, which include broker-dealers, "are increasingly using third-party vendors to perform a wide range of core business and regulatory oversight functions," a trend that has accelerated with the COVID-19 pandemic.[680] One report describes the results of a 2022 survey of 248 advisers and independent broker-dealers.[681] The survey found that 32 percent of the registered investment advisers and 50 percent of the independent broker-dealers that responded to the survey reported outsourcing investment management functions, and that while these proportions had not changed significantly in the past decade, half of the respondents who do outsource some of these functions reported an increase in their use of service providers. In addition, a different recent report finds that 33 percent of asset managers surveyed outsource their entire back-office function and 20 percent outsource their entire middle-

677 *See* IAA Comment Letter 1.

678 In addition, in response to this commenter, we have added further details on the current regulatory framework, in particular with respect to the obligations of covered institutions regarding their service providers and the notification obligations of service providers. *See supra* section IV.C.2. Also, we have supplemented the analysis of the benefits and costs of the final amendments' service provider requirements. *See infra* section IV.D.1.c. The supplemental review described here is designed to help us analyze and respond to commenters, and also to provide additional context for this analysis.

679 Potential service providers include a wide range of firms fulfilling a variety of functions. The internal organization of covered institutions, including their reliance on service providers, is not generally publicly observable. Although certain regulatory filings shed a limited light on the use of third-party service providers (e.g., transfer agents' reliance on third parties for certain functions and investment advisers' reliance on third parties for recordkeeping), we are unaware of any data sources that provide detail on the reliance of covered institutions on service providers.

680 *See* FINRA, *Regulatory Notice 21-29, supra* footnote 515.

681 *See* FlexShares, *The Race to Scalability 2022* (July 2022).

office function.[682] By the nature of their business models, most of the operations of investment companies are carried out by service providers.[683] Finally, many transfer agents outsource many functions.[684] Hence, all types of covered institutions affected by the final amendments commonly retain service providers to some extent.

D. Benefits and Costs of the Final Rule Amendments

The final amendments can be divided into four main components. First, they create a requirement for covered institutions to adopt policies and procedures for the protection of customer information. The policies and procedures must include an incident response program to address unauthorized access to or use of customer information, including by providing notification to individuals affected by an incident during which their sensitive customer information was, or is reasonably likely to have been, accessed or used without authorization. The response program must also include the establishment, maintenance, and enforcement of written policies and procedures reasonably designed to require oversight of service providers, including to ensure service providers take appropriate measures to protect against unauthorized access to or use of customer information. Second, the amendments define the information covered by the safeguards rule and the disposal rule,[685] and extend the application of the safeguards rule to transfer agents. Third, the amendments require covered institutions (other than

---

682 *See* Cerulli Associates, *Asset Managers Turn to Outsourcing Providers for Operating Model Sustainability* (Nov. 22, 2022), *available at* https://www.cerulli.com/press-releases/asset-managers-turn-to-outsourcing-providers-for-operating-model-sustainability ("Cerulli Report").

683 *See* Investment Company Institute, *How US-Registered Investment Companies Operate and the Core Principles Underlying Their Regulation* (May 2022), available at https://www.ici.org/system/files/2023-06/us-reg-funds-principles.pdf.

684 *See supra* footnote 670. *See also Interagency Guidance on Third-Party Relationships: Risk Management*, 88 FR 37920, 37937 (June 9, 2023), which may cover some transfer agents registered with a regulatory agency other than the Commission.

685 *See* final rule 248.30(a)(1), 248.30(b), 248.30(d)(1), and 248.30(d)(5).

funding portals) to maintain and retain records documenting compliance with the amended rules.[686] Fourth, they incorporate into regulation an existing statutory exemption for annual privacy notices. Below we discuss the benefits and the costs of each component in turn.

Some commenters criticized, generally, the discussion of benefits and costs in the Proposing Release. One commenter stated that the Commission should "undertake a more expansive, accurate, and quantifiable assessment of the specific and cumulative costs, burdens, and economic effects that would be placed on investment advisers by the proposed requirements, as well as of the potential unintended consequences for their clients."[687] Another commenter stated a need for more in-depth analysis of how the proposed amendments might impact transfer agents, their customers (issuers of securities), and securityholders.[688] Other commenters did not directly disagree with the analysis in the Proposing Release, but stated that the proposed amendments would place a high overall burden on covered institutions, including smaller institutions.[689]

In response to these commenters, we have supplemented the analysis of the benefits and the costs of the final amendments regarding the timing requirement for notification of customers affected by a breach, including by providing more details on how the requirements differ from the baseline;[690] different elements required to be included in a notice to affected individuals;[691]

---

686 As discussed above, funding portals are not subject to the recordkeeping obligations found under Rule 17a-4. Funding portals are instead obligated, pursuant to Rule 404 of Regulation Crowdfunding, to make and preserve all records required to demonstrate their compliance with Regulation S-P for five years, the first two years in an easily accessible place. *See supra* footnote 385; *see also* 17 CFR 227.404(a)(5).

687 *See* IAA Comment Letter 1.

688 *See* STA Comment Letter 2.

689 *See, e.g.*, SIFMA Comment Letter 2; ASA Comment Letter.

690 *See infra* section IV.D.1.b(2); *see also supra* section IV.C.2.a(2).

691 *See infra* section IV.D.1.b(5); *see also supra* section IV.C.2.a(2).